

The IS-LM Model: Extensions and Applications

Intermediate Macroeconomics

Fall 2025, Sciences Po Campus du Havre

Lecture 9

Announcements

- Quiz 3 announced today, due 11/11
- Lecture 8: We introduced the IS-LM model
- Today, we extend the model: real interest rates & risk premium
- Today, application: 2008 Financial Crisis
- Book: 6-1, 6-4 and 6-5
(We will skip 6-2 and 6-3)

Nominal vs. Real Interest Rates

T-Bills

Terminology:

- **Treasury bill (T-Bill):** A U.S. government bond with a maturity of up to 1 year.
- **Treasury note:** Maturity between 1 and 10 years
- **Treasury bond:** Maturity of more than 10 years

Some practicalities:

- T-bills are issued by the U.S. government. When you buy one, you are essentially lending money to the government.
- T-bills have a **face value** of a certain amount, which is what they are actually worth.
- T-bills are sold for less – we call it “**at a discount**”
 - Example: T-bill may be worth \$10,000, but you would buy it for \$9,600

T-Bills

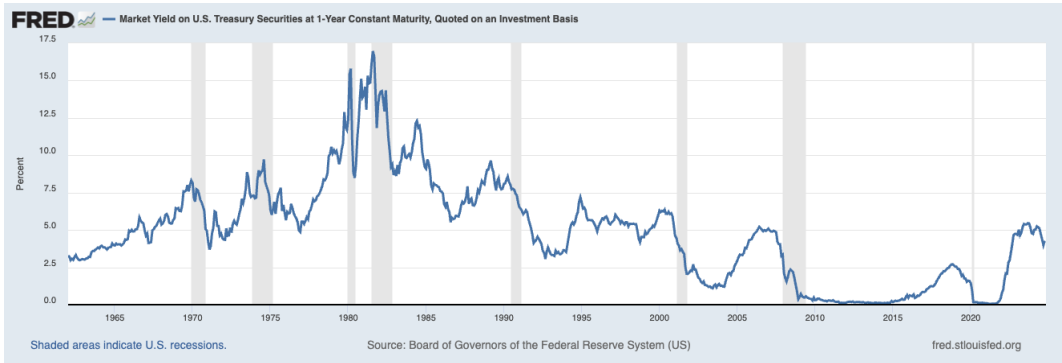
- T-bills bills are one of the safest forms of investment in the world because they are backed by the U.S. government.
- They are considered risk-free.
- T-bills were first used in the United States during World War I, as a source of emergency funds to help balance the unprecedentedly high public debt.
- By the end of World War II, T-bills had become the most popular form of short-term government security.

1 Year Treasury Rate

The current 1 year treasury yield as of October 2024 is 4.24%.

The borrowing in 1980s was much more expensive for the government.

Why? Inflation



Nominal vs. Real Interest Rates

- **Nominal interest rate** is the interest rate in terms of dollars.
- **Real interest rate** i is the interest rate in terms of a basket of goods.
We must adjust the nominal interest rate to take into account expected inflation.

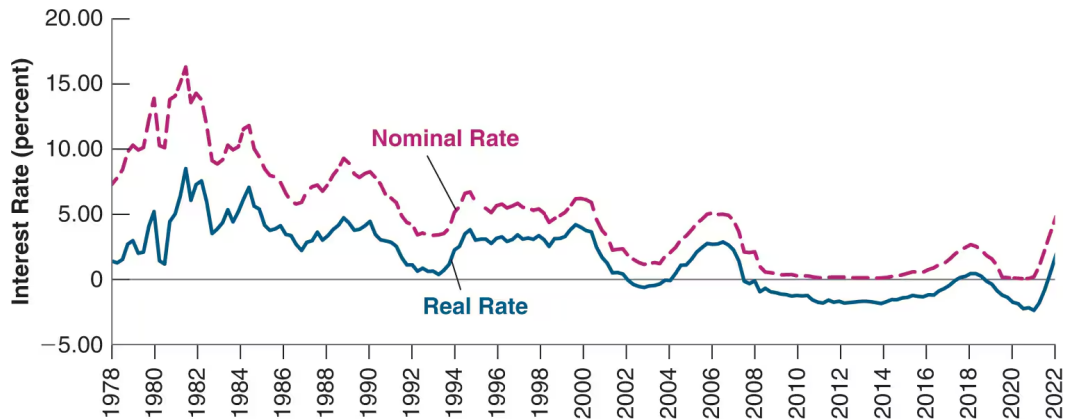
$$r_t \approx i_t - \pi_{t+1}^e$$

Nominal vs. Real Interest Rates

$$r_t \approx i_t - \pi_{t+1}^e$$

- When expected inflation equals zero, the nominal interest rate and the real interest rate are equal
- Because expected inflation is typically positive, the real interest rate is typically lower than the nominal interest rate
- For a given nominal interest rate, the higher expected inflation, the lower the real interest rate

Nominal vs. Real Interest Rates



Risk and Return Premia

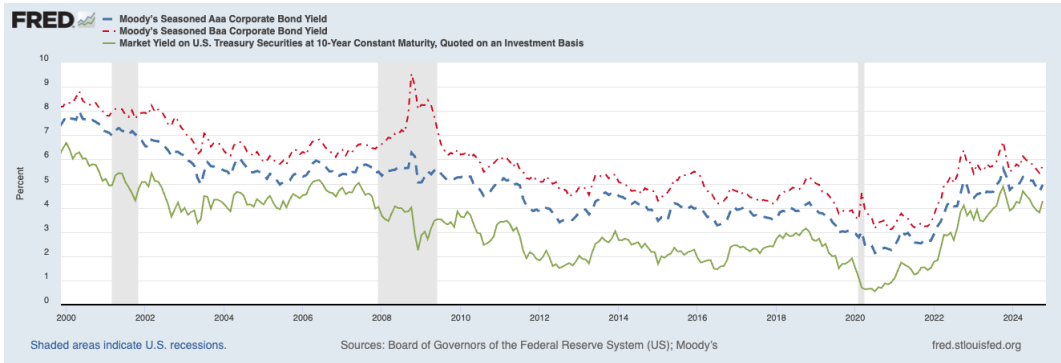
Differences across bonds

- Until now, we assumed that all bonds are the same
- In reality, bonds differ:
 - Maturity (1Y vs 10Y)
 - Default risk (sovereign vs corporate bonds)
- Some bonds are risky, so bond holders require a **risk premium** (x)
- Risk premia are determined by:
 - **The probability of default**

$$\underbrace{(1 + i)}_{\text{return of the riskless bond}} = \underbrace{(1 - p)}_{\text{probability of no default}} \underbrace{(1 + i + x)}_{\text{return if no default}} + \underbrace{p}_{\text{probability of default}} \underbrace{(0)}_{\text{return if default}}$$

- The degree of **risk aversion** of bond holders

Risk and Return Premia



Extending the IS-LM

Extending the IS-LM

We extend the IS-LM to reflect the distinction between

1. the nominal interest rate and the real interest rate
2. the policy rate set by the central bank and the interest rates faced by borrowers

IS relation

$$Y = C(Y - T) + I(Y, i - \pi^e + x) + G$$

$$Y = C(Y - T) + I(Y, r + x) + G$$

Why?

- r : investment depends on the real interest rate rather than nominal
- x : risk premium captures the probability of default and the risk aversion

Extending the IS-LM

IS relation

$$Y = C(Y - T) + I(Y, r + x) + G$$

LM relation

$$r = \bar{r}$$

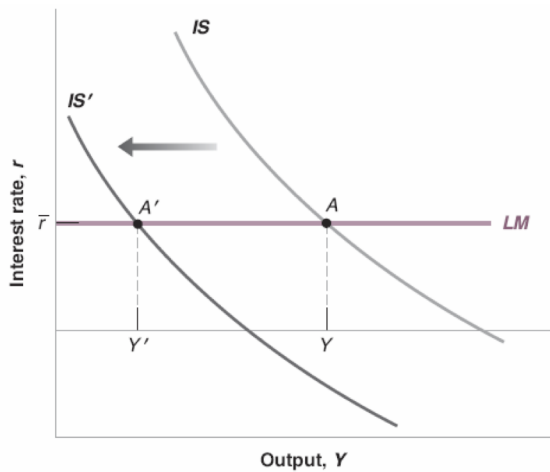
Why?

- The central bank now chooses the real policy rate r
- r also enters the IS equation as part of the borrowing rate $(r + x)$ for consumers and firms:

Extending the IS-LM: Financial Shock

Risk premium increases (x)

An increase in x leads to a shift of the IS curve to the left and a decrease in equilibrium output.



Financial Shock and Policy Options

- In our previous example: $\uparrow x \implies$ IS shifts to the left $\implies \downarrow$ equilibrium output
- What can policy-makers do to address it?

1. Fiscal expansion

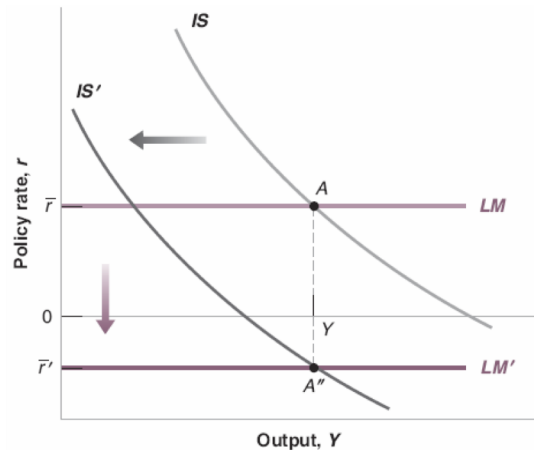
- “Shift the IS back”: $\uparrow G$ or $\downarrow T$
- This might require a very large cut in taxes / massive spending
- Given that the shock came from investors (financial sector): government might be reluctant to do something

2. Monetary expansion

- Aggressive monetary policy: large $\downarrow r$
- If sufficiently large, a decrease in the policy rate can in principle offset the increase in the risk premium.
- The zero lower bound may however put a limit on the decrease in the real policy rate.

Financial Shock and Monetary Policy

- A sufficiently large MoPo response can in principle offset the increase in the risk premium
- The zero lower bound may however put a limit on the decrease in the real policy rate



2008 Financial Crisis

Modern banking

- Narrow banking, commercial banks as intermediaries to receive deposits and lend them to borrowers
- In the modern financial market, there are “nonbanks”
 - i.e. mortgage companies, money market funds, and hedge funds etc.
 - The non-bank part of the financial system is called shadow banking

Modern Banking

- Bank directly invested 20 of their own fund
 - Bank capital = Assets – Liabilities
- The liabilities of the bank comprise not only deposits but also short - term borrowing from the investors and other banks.
- Using their own funding (capital) and liabilities (short-term debt), the bank purchased 100 Assets (long-term maturity)
 - the spread for profit

Bank Balance Sheet

Assets 100	Liabilities 80
	Capital 20

$$\text{Capital ratio} = \text{Capital} / \text{Assets} \\ = 20/100 = 0.2$$

$$\text{Leverage ratio} = \text{Assets} / \text{Capital} \\ = 100/20 = 5$$

From a Housing Problem to a Financial Crisis

The increase in housing prices from 2000 to 2006 was followed by a sharp decline thereafter.



Modern banking

- The previous chapter explained the bank run due to the maturity mismatch
- Modern banking has a different liquidity composition
 - Banks do short-term borrowing from other banks or investors
 - Short-term borrowings are not covered by the federal deposit insurance
 - Liquidity provision by CB faces difficulties in assessing which banks are insolvent or not
- Banks are required to meet the capital adequate ratio (capital over asset ratio) to prevent
 - Still non-bank financial institution can get away from this regulation by offshoring its business from the bank balance sheet (i.e. structural investment vehicle)

From a Housing Problem to a Financial Crisis

- The 2000s were a period of unusually **low interest rates**, which stimulated housing demand.
- Mortgage lenders was increasingly willing to make loans to risky borrowers with **subprime mortgages**
- From 2006 on, many home mortgages went underwater (when the value of the mortgage exceeded the value of the house).
- Lenders faced large losses as many borrowers defaulted.

Securitization

Securitization is the creation of securities based on a bundle of assets, such as mortgage-based securities (MBS).

- Senior securities have first claims on the return from the bundle of assets
- Junior securities come after.
- Example: collateralized debt obligations (CDOs)

Securitization and Wholesale funding

- Banks did not issue MBS directly. They avoided financial regulations with structured investment vehicles (SIVs) .
- Wholesale funding is a process in which banks rely on borrowing from other banks or investors to finance the purchase of their assets.
- In 2000s, SIVs were entirely funded through wholesale funding.
- Wholesale funding resulted in liquid liabilities.

From a Housing Problem to a Financial Crisis

The financial crisis led to a sharp drop in confidence, which bottomed in early 2009.



From a Housing Problem to a Financial Crisis

- The demand for goods decreased due to the high cost of borrowing, lower stock prices, and lower confidence.
- The IS curve shift to the left.
- Policy makers responded to this large decrease in demand.

Financial Policies

- Federal deposit insurance was raised from \$100,000 to \$250,000,
- The Fed provided widespread liquidity to the financial system through liquidity facilities, and increased the number the assets that could serve as collateral,
- The government introduced the Troubled Asset Relief Program (TARP)

Monetary and Fiscal Policies

Monetary Policy

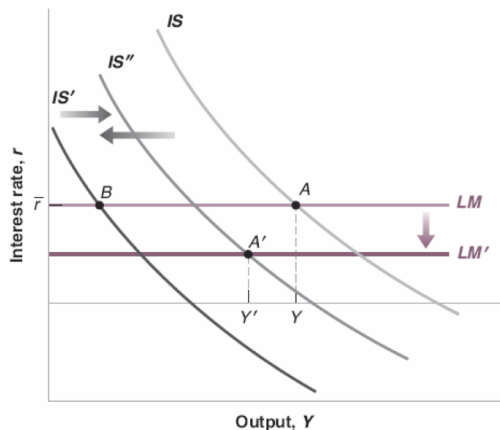
- The federal funds rate was down to zero by December 2008.
- The Fed also used unconventional monetary policy, which involved buying other assets as to directly affect the rate faced by borrowers.

Fiscal Policy

- The American Recovery and Reinvestment Act was passed in February 2009, calling for \$780 billion in tax reductions and spending increases

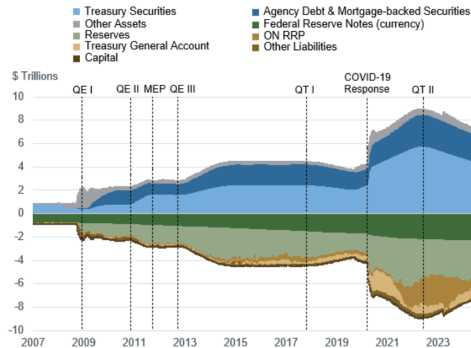
Policy response

- The financial crisis led to a shift of the IS to the left.
- Financial and fiscal policies led to some shift back to the IS to the right.
- Monetary policy led to a shift of the LM curve down.
- Policies were not enough however to avoid a major recession.



Unconventional Monetary Policy

Federal Reserve Assets and Liabilities



Note: "QE" is quantitative easing (increase of securities holdings), "MEP" is Maturity Extension Program, "QT" is quantitative tightening (reduction of securities holdings).
Source: Board of Governors of the Federal Reserve System

Figure: Large Scale Asset Purchase by Feds